



Modelling Financial Markets

Seminar 15: LLMs and Sentiment Analysis — Hands-on



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Acronyms used in this material

- ▣ AI: Artificial Intelligence
- ▣ AUC: Area Under the (ROC) Curve
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CAR: Cumulative Abnormal Return
- ▣ CI: Confidence Interval
- ▣ ECE: Expected Calibration Error
- ▣ FNSPID: Financial News and Stock Price Integration Dataset
- ▣ GPT-4: Generative Pre-trained Transformer 4
- ▣ LLM: Large Language Model
- ▣ LM: Loughran-McDonald (financial sentiment dictionary)
- ▣ NLL: Negative Log-Likelihood
- ▣ OLS: Ordinary Least Squares
- ▣ PPI: Prediction-Powered Inference
- ▣ SD: Standard Deviation
- ▣ SE: Standard Error
- ▣ TF-IDF: Term Frequency - Inverse Document Frequency
- ▣ US: United States
- ▣ UTC: Coordinated Universal Time (Temps Universel Coordonné)

Today's Question and Required Outputs

- ▣ Question of the seminar: **can a language model read financial sentiment, and does that sentiment predict returns one can still trade?**
- ▣ Route
 - ▶ A1, A3, A7 on paper: how a misclassified label attenuates a slope, how precise a paired accuracy test is, at what cost a strategy stops earning
 - ▶ B1: is FinBERT better than a dictionary? B5: do headlines predict returns or describe them? B7: does the LLM remember the market?
 - ▶ C2: find the three errors in an AI answer; B2 at home
- ▣ Required outputs (checklist on the last core slide): the A answers, one table and one figure per B exercise, the C2 corrections, one interpretation sentence per exercise
- ▣ Notebook for today: [open the seminar notebook in Colab](#); every exercise has a section with the same label
- ▣ Seminar: Antoaneta Amza, PhD candidate, antoaneta.amza@csie.ase.ro

Exercise Map

Exercise	Question	Type	Model
A1	how much does a misclassified dummy shrink a slope?	Solved	–
A3	how precise, and how powerful, is a paired accuracy test?	Solved	–
A7	at what cost per trade does the strategy stop earning?	Solved	–
B1	is FinBERT more accurate than the LM dictionary?	Solved	–
B5	do headlines predict returns, or describe them?	Solved	–
B7	does the LLM remember the market?	Solved	–
C2	what is wrong in an AI answer?	Proposed	B1, B5, B7
B2	is a 7-billion-parameter LLM better than FinBERT? (homework)	Proposed	B1
Extension packs	A2, A4, A5, A6, A8, B3, B4, B6, B8–B11, C1, C3 (optional, after the seminar)	both	named on each task

- ☐ **[Solved]**: full solution in the slides and in the notebook, a model to copy; **[Proposed]**: you solve it, the solution is discussed in class
- ☐ Part A: derivations on paper; Part B: verification and inference on real data, each ending with an interpretation question; Part C: open questions and AI critique

Data Used

Data	Content	Saved scores	Used in
Twitter Financial News (validation set, Hugging Face Hub)	2 388 headlines from 2022: 1 566 neutral, 475 positive, 347 negative	LM, FinBERT, Qwen2.5 (P1–P3), MiniLM	A1–A5, A8, B1–B4, B8, B9, C2
Financial PhraseBank Malo et al. (2014)	4 846 news sentences, 2 264 with full agreement	same scores	B3
FNSPID Dong, Fan & Peng (2024)	106 820 headlines on 16 US stocks, 27.10.2009–18.12.2023	daily LM, FinBERT, Qwen2.5-7B scores	A7, A8, B5, B6, B10, B11, C1–C3
Stocks and SPY	adjusted daily prices of the course data, one series per asset (field used: adjusted close), 2009–2023	–	B5, B6, C1, C3
S&P 500	monthly closes, Jan 2000 – Aug 2026	Qwen2.5 answers about each month	B7, C2

- ▣ The scores are saved with the chapter's Quantlets, [Quantlets/Ch_15](#): no language model has to run in class
- ▣ C3 (extension pack) needs a new scoring run of the FNSPID headlines

Setup: Run and Verify

```
# notebook section "Setup" defines the functions that
# read the data
pb, tw = eval_texts()    # PhraseBank, Twitter
P, rets = news_panel()  # FNSPID stock-days
print(len(pb), len(tw))
print(tw['label'].value_counts())
print(P.shape,
      P.index.get_level_values('ticker').nunique())
```

- Expected output
 - ▶ 4 846 and 2 388 rows
 - ▶ neutral 1 566, positive 475, negative 347
 - ▶ (18 282, 19) and 16 stocks
- Each row of tw holds the text, the human label and the probabilities of every model (e.g. finbert_negative, qwen7B_positive)
- If your counts differ, stop and check the files before any exercise

Notation Used in Every Exercise

- Dictionary tone $\tau = (P - N)/(P + N)$, with P , N the counts of positive and negative words; LM: Loughran–McDonald financial lists [Loughran & McDonald \(2011\)](#)
- Model probabilities $p_k = e^{z_k/T} / \sum_j e^{z_j/T}$ (z_k : logit of answer k , T : temperature); label $\arg \max_k p_k$; net score $s = p_{\text{pos}} - p_{\text{neg}}$; FinBERT [Araci \(2019\)](#), Qwen2.5 [Qwen Team \(2024\)](#)
- Two classifiers A and B on the same texts: n_{01} = A wrong and B right, n_{10} = A right and B wrong; McNemar test [McNemar \(1947\)](#): under H_0 , $n_{01} \sim \text{Bin}(n_{01} + n_{10}, 0.5)$
- Day d : the first trading day on or after the headline date; $d + 2$: the first return surely after publication; h : days after d
- bp: basis point, 0.01%; SE: standard error; CI: 95% confidence interval; t_{NW} : Newey–West t [Newey & West \(1987\)](#); clustered SE: [Petersen \(2009\)](#)

A1 [Solved]: Attenuation from Misclassification — Task

- **Question:** a regression uses the FinBERT label "negative" instead of the truth; by how much is the slope shrunk?
- **Data**
 - ▶ Twitter Financial News, validation set: $N = 2\,388$ headlines from 2022, each labelled negative, neutral or positive; the human label is treated as the truth
 - ▶ Model $r = a + \beta s^* + \varepsilon$, $s^* = 1$ if the headline is truly negative; we observe only $\hat{s} = 1$ when FinBERT predicts negative
 - ▶ Non-differential error: given s^* , $\hat{s}$ says nothing about r , so $\text{Cov}(\varepsilon, \hat{s}) = 0$
 - ▶ $\pi = P(s^* = 1)$, $p = P(\hat{s} = 1)$, false alarms
 $\alpha_0 = P(\hat{s} = 1 \mid s^* = 0)$, misses $\alpha_1 = P(\hat{s} = 0 \mid s^* = 1)$
- **Tasks**
 - (a) Show that $\text{plim } \hat{b} = \beta \text{Cov}(s^*, \hat{s}) / \text{Var}(\hat{s})$.
 - (b) Derive $\lambda = \pi(1 - \pi)(1 - \alpha_0 - \alpha_1) / [p(1 - p)]$.
 - (c) Compute π , p , α_0 , α_1 from the matrix, then λ to three decimals.
 - (d) Compute $\text{Cov}(u, s^*)$ for $u = \hat{s} - s^*$: is the error classical?
- **Report:** λ , $1/\lambda$ and the sign of $\text{Cov}(u, s^*)$
- **Interpretation:** what share of a true effect survives in a slope on the FinBERT dummy?

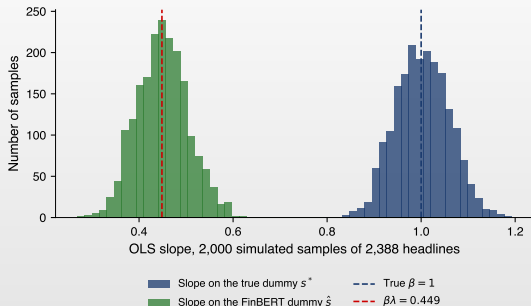
FinBERT confusion matrix, Twitter

True \ predicted	neg	neu	pos	Total
neg	264	71	12	347
neu	213	1 184	169	1 566
pos	58	133	284	475
Total	535	1 388	465	2 388

A1 [Solved]: Solution (1/2)

- (a) For one regressor $\text{plim } \hat{b} = \text{Cov}(r, \hat{s}) / \text{Var}(\hat{s})$ and $\text{Cov}(r, \hat{s}) = \beta \text{Cov}(s^*, \hat{s}) + \text{Cov}(\varepsilon, \hat{s})$; the last term is 0 by the non-differential assumption
- (b) Both variables are binary: $\text{Cov}(s^*, \hat{s}) = P(s^* = 1, \hat{s} = 1) - \pi p = \pi(1 - \alpha_1) - \pi p$
- Total probability: $p = \pi(1 - \alpha_1) + (1 - \pi)\alpha_0$; substituting, $\text{Cov}(s^*, \hat{s}) = \pi(1 - \pi)(1 - \alpha_0 - \alpha_1)$
- $\hat{s}$ is Bernoulli, so $\text{Var}(\hat{s}) = p(1 - p)$; the ratio of the two is λ

A1 [Solved]: Solution (2/2)

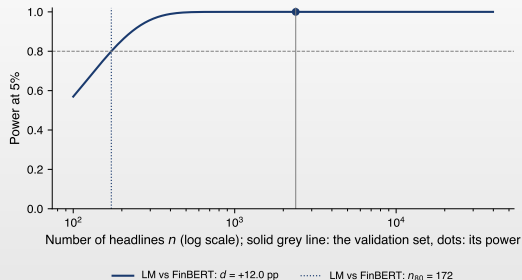


- (c) $\pi = 347/2\,388 = 0.1453$, $p = 535/2\,388 = 0.2240$, $\alpha_1 = 1 - 264/347 = 0.2392$, $\alpha_0 = 271/2\,041 = 0.1328$
- $\lambda = \frac{0.1242 \times 0.6280}{0.2240 \times (1 - 0.2240)} = 0.449$, so $\beta = 2.23 \hat{\beta}$
- (d) $\text{Cov}(u, s^*) = 0.0780 - 0.1242 = -0.0462 < 0$: the error is negatively correlated with the truth, not classical [Aigner \(1973\)](#)
- Chart: 2 000 simulated samples with these rates and $\beta = 1$; the slope on $\hat{s}$ averages 0.450 (SD 0.054), as λ predicts
- Interpretation: a slope on the FinBERT dummy keeps only about 45% of the true effect

A3 [Solved]: Paired Accuracy Difference: Variance, Exact Test, Power — Task

- **Question:** how precise is the accuracy gap between two classifiers scored on the same headlines, and how many headlines does it need?
- **Inputs**
 - ▶ Twitter validation set, $n = 2\,388$ independent headlines; A = LM dictionary, B = FinBERT
 - ▶ Discordant counts: $n_{01} = 539$ (A wrong, B right), $n_{10} = 252$ (A right, B wrong)
 - ▶ $D_i = 1$ if A wrong and B right, -1 if A right and B wrong, 0 otherwise; $d = ED_i = \text{acc}_B - \text{acc}_A$, $q = P(D_i \neq 0)$
- **Tasks**
 - (a) Show $\hat{d} = (n_{01} - n_{10})/n$ and derive $\text{Var}(\hat{d}) = (q - d^2)/n$.
 - (b) Compute $\hat{d}$, SE, the 95% CI and $z = \hat{d}/\text{SE}$.
 - (c) Run the exact McNemar test at 5% and state the decision.
 - (d) Compute the power $\Phi(|d|\sqrt{n}/\sqrt{q - d^2} - 1.96)$ if the true d is $\hat{d}$, and $n_{80} = (1.96 + 0.84)^2(q - d^2)/d^2$.
- **Report:** $\hat{d}$, SE, CI, z , exact p , the decision, the power and n_{80}
- **Interpretation:** which headlines determine the variance of $\hat{d}$?

A3 [Solved]: Solution

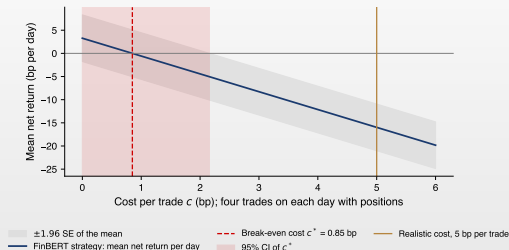


- (a) $\hat{d} = \frac{1}{n} \sum_i D_i = (n_{01} - n_{10})/n$; $D_i^2 = 1$ exactly when one is right and the other wrong, so $ED_i^2 = q$, $\text{Var}(D_i) = q - d^2$ and $\text{Var}(\hat{d}) = (q - d^2)/n$
- (b) $\hat{d} = (539 - 252)/2388 = 0.1202$,
 $\hat{q} = 791/2388 = 0.3312$, $\text{SE} = 1.15$ pp, $\text{CI} [9.76; 14.28]$ pp, $z = 10.43$
- (c) Under H_0 we expect 395.5 on each side; exact $p < 0.001$: reject equal accuracy
- (d) Power $\Phi(+8.47) \approx 100\%$;
 $n_{80} = 7.84 \times (0.3312 - 0.01444)/0.01444 \approx 170$ headlines (chart)
- Interpretation: only the discordant headlines (q) enter the variance; a 12-point gap is detectable with a few hundred headlines

A7 [Solved]: Break-Even Cost with a Standard Error — Task

- **Question:** at what cost per trade does the FinBERT strategy stop earning, and how precisely do we know that cost?
- **Inputs**
 - ▶ Strategy of Lecture 15: at the close of $d + 1$ take $\text{sign}(s_{i,d})$ in every stock with news on day d , hedged with SPY, closed one day later; four trades on each day with positions; 3 568 trading days
 - ▶ FinBERT signal: mean gross return over all days $\hat{\mu} = 3.27$ bp, $t_{NW} = 1.27$ ($L = 8$ lags), share of days with positions $a = 0.962$, gross Sharpe ratio $\sqrt{252} \hat{\mu} / \sigma = 0.36$
- **Tasks**
 - (a) Write the mean net daily return and the net Sharpe ratio as functions of the cost c per trade.
 - (b) Derive the break-even cost c^* at which the mean net return is zero.
 - (c) Compute $\text{SE}(\hat{\mu}) = \hat{\mu} / t$, then $\text{SE}(c^*)$ by the delta method and the 95% CI of c^* .
 - (d) Recover σ from the gross Sharpe ratio and compute the net Sharpe ratio at $c = 5$ bp.
- **Report:** c^* , its SE and CI, the net Sharpe ratio
- **Interpretation:** is a realistic cost of 5 bp per trade inside the CI of c^* ?

A7 [Solved]: Solution



- (a) On a share a of days we pay $4c$: $\mu_{\text{net}} = \mu - 4ac$,
 $SR_{\text{net}} = \sqrt{252} (\mu - 4ac) / \sigma$, with σ barely changed by costs
- (b) $c^* = \mu / (4a) = 3.27 / 3.85 = 0.85$ bp per trade
- (c) $SE(\hat{\mu}) = 3.27 / 1.27 = 2.57$ bp; with a fixed by the news calendar, $SE(c^*) = 2.57 / 3.85 = 0.67$ bp, CI $[-0.46; 2.16]$ bp
- (d) $\sigma = \sqrt{252} \times 3.27 / 0.36 = 143$ bp;
 $SR_{\text{net}} = \sqrt{252} (3.27 - 3.85 \times 5) / 143 = -1.77$
- Interpretation: 5 bp lies far above the CI of c^* ; with daily turnover the strategy cannot survive realistic costs

 MFM_ch15_sem_partA

B1 [Solved]: Is FinBERT Better than LM? — Task

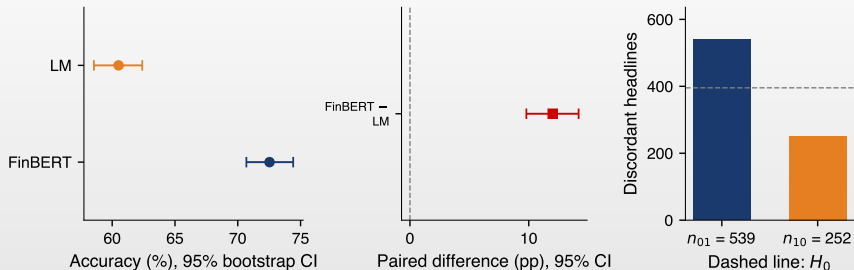
- **Question:** is the fine-tuned FinBERT more accurate than the Loughran–McDonald dictionary on headlines it was not trained on?
- **Data**
 - ▶ Twitter Financial News, validation set: $N = 2\,388$ headlines from 2022, each labelled negative, neutral or positive; the human label is treated as the truth
 - ▶ A = LM dictionary (label = sign of the tone), B = FinBERT (label = largest probability); notebook function `b_pairs`
- **Tasks**
 - (a) Compute each accuracy with a percentile bootstrap 95% CI: resample the headlines 2 000 times (seed 0).
 - (b) Compute the difference $B - A$ with a paired bootstrap CI: the same resampled headlines for both classifiers.
 - (c) Count n_{01} , n_{10} ; run the exact McNemar test and $(n_{01} - n_{10})^2 / (n_{01} + n_{10})$ against χ^2_1 (3.84).
- **Report:** a table of the two accuracies, their CIs and the difference; n_{01} , n_{10} , both p -values and the decision
- **Interpretation:** why is the paired test more powerful than comparing the two intervals, and does the gap matter for trading?

B1 [Solved]: Solution (1/2)

	Accuracy (%)	CI 95%
LM	60.5	[58.5; 62.4]
FinBERT	72.5	[70.7; 74.4]
Difference (pp)	+12.0	[+9.8; +14.2]

- Step 1: labels of both classifiers on the 2 388 headlines; step 2: 2 000 bootstrap resamples of the headline indices; step 3: both accuracies and their difference in each resample
- (a)–(b) The two intervals do not overlap and the paired CI of the difference excludes 0; it agrees with the analytic CI of A3

B1 [Solved]: Solution (2/2)

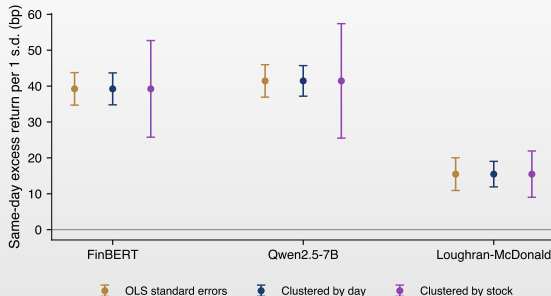


- (c) $n_{01} = 539$, $n_{10} = 252$, 395.5 expected under H_0 ; exact $p < 0.001$; $\chi^2 = 104.1 > 3.84$: reject equal accuracy
- Pairing removes the variation in headline difficulty: headlines both get right, or both get wrong, carry no information
- Interpretation: FinBERT is clearly better than the dictionary outside its training data; a 12-point gain in labelling says nothing yet about returns (B5)

B5 [Solved]: Reaction or Prediction, and Which Standard Errors? — Task

- **Question:** does the daily headline score explain the return of the news day, or forecast the first return one can still trade, and how wide are the honest intervals?
- **Data**
 - ▶ FNSPID panel: 18 282 stock-days with news, 16 US stocks, 27.10.2009–18.12.2023; $z_{i,d}$: daily net score, standardised over the panel; r^e : excess return over SPY (%), from the adjusted daily closes of the course data
 - ▶ Scores: FinBERT, Qwen2.5-7B, LM
- **Tasks**
 - (a) Estimate $r_{i,d+h}^e = a + b z_{i,d} + \varepsilon$ by OLS for the three scores and $h = 0, 1, 2$; report b in bp per one SD of the score.
 - (b) Compute three SEs: OLS, clustered by day ($G = 3\,434$), clustered by stock ($G = 16$), with the small-sample factor $\frac{G}{G-1} \frac{n-1}{n-2}$.
 - (c) Compare each t with 1.96, and the stock-clustered t with the t_{15} critical value 2.13.
- **Report:** a chart of the $h = 0$ slopes with the three intervals, and a table of the regressions
- **Interpretation:** why do $h = 0$ and $h = 2$ differ, and can a clustered SE with 16 clusters be trusted?

B5 [Solved]: Solution (1/2)



- Same-day slopes with the 95% intervals of the three standard errors: the stock-clustered interval is the widest
- Clustering by day is close to OLS: with only 5.3 stocks per day there is little common noise
- Clustering by stock multiplies the FinBERT SE at $h = 0$ by 3.0: the slope differs across stocks and each stock's errors are correlated over time

B5 [Solved]: Solution (2/2)

Score	h	b (bp)	SE OLS	SE day	SE stock	t day	t stock
FinBERT	0	+39.2	2.3	2.3	6.9	+17.27	+5.71
FinBERT	2	-0.5	2.2	2.2	1.5	-0.25	-0.35
Qwen2.5-7B	0	+41.4	2.3	2.2	8.1	+19.07	+5.10
Qwen2.5-7B	2	+2.3	2.2	1.9	1.0	+1.19	+2.41
LM	0	+15.5	2.3	1.8	3.3	+8.53	+4.70
LM	2	-0.0	2.2	1.7	0.9	-0.01	-0.03

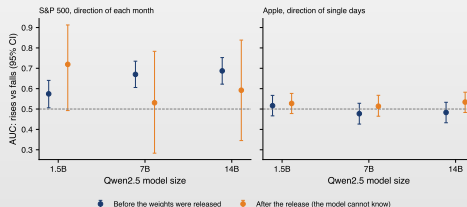
- $h = 0$: large under every SE (t by stock above 4.7): headlines describe the day's move, often the move itself ("shares jump")
- $h = 2$: FinBERT -0.5 bp ($t = -0.25$ by day); Qwen $t = +2.41$ by stock, but $p = 0.061$ by wild cluster bootstrap [Cameron, Gelbach & Miller \(2008\)](#) and 0.37 after Holm over nine regressions (extension B10)
- Interpretation: sentiment explains the news day, not the tradable days; with 16 stocks, inference needs few-cluster methods

B7 [Solved]: Does the Model Remember the Market? — Task

- **Question:** asked without any data, can Qwen2.5 tell the months in which the S&P 500 rose from those in which it fell?
- **Data**
 - ▶ S&P 500 monthly closes (GSPC.INDX), January 2000 to August 2026; a month is "up" if its return is positive
 - ▶ Qwen2.5 1.5B, 7B, 14B (weights published 19 September 2024) asked "Did the S&P 500 index rise or fall during ⟨month year⟩?"; score $P(\text{rise})$
 - ▶ Before: 296 months to August 2024 (185 up); after: 23 months from October 2024 (13 up)
- **Tasks**
 - (a) Compute the $\text{AUC} = U / (n_{\text{up}} n_{\text{down}})$ (U : Mann–Whitney) per model and period, with a 2000-draw bootstrap CI over months.
 - (b) Permutation test for 14B before the release: shuffle the up/down labels 2000 times.
 - (c) Test $\text{AUC}_{\text{before}} = \text{AUC}_{\text{after}}$ with DeLong variances [DeLong et al. \(1988\)](#) and find the smallest drop detectable with 80% power.
- **Report:** the AUC table with CIs, the permutation p , the DeLong z and p , the detectable drop
- **Interpretation:** what does an AUC above 0.5 mean when the model sees no data?

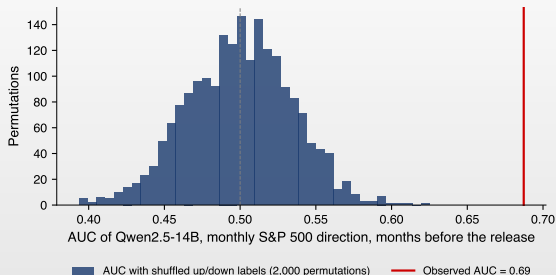
B7 [Solved]: Solution (1/2)

Qwen2.5	AUC before	CI 95%	AUC after	CI 95%
1.5B	0.57	[0.51; 0.64]	0.72	[0.49; 0.91]
7B	0.67	[0.61; 0.73]	0.53	[0.28; 0.78]
14B	0.69	[0.62; 0.75]	0.59	[0.35; 0.84]



- (a) 7B and 14B rank up months above down months far above chance before the release; 1.5B barely; after the release all intervals contain 0.5

B7 [Solved]: Solution (2/2)



- (b) Permutation p for 14B before the release < 0.001 : no shuffled AUC reaches the observed one (chart)
- (c) DeLong: SE before 0.032, after 0.126; $z = 0.73$, $p = 0.47$: do not reject equality; detectable drop
 $2.80 \sqrt{SE_{\text{before}}^2 + SE_0^2} = 0.36$, about 101 post-release months needed
- Interpretation: without data, the discrimination comes from the training text; a backtest on pre-release news mixes forecasting with remembering [Glasserman & Lin \(2023\)](#)

C2 [Proposed]: Critique an AI Answer — the Answer

- **Student prompt:** “I have FNSPID headlines for 16 US stocks, 2009–2023, and daily excess returns vs SPY. Check that Qwen2.5-7B reads sentiment, then test whether its headline sentiment forecasts returns out of sample.”
- **Code of the AI assistant** (qwen_label, qwen_score: the zero-shot scorer of Lecture 15)

```
import pandas as pd, statsmodels.formula.api as smf
from sklearn.metrics import accuracy_score
tw['pred'] = [qwen_label(t) for t in tw['text']] # Twitter validation set
acc = accuracy_score(tw['label'], tw['pred'])
news['day'] = first_trading_day_on_or_after(news['date'])
news['s'] = [qwen_score(t) for t in news['title']] # P(pos) - P(neg)
d = news.groupby(['ticker', 'day'])['s'].mean().reset_index()
d['z'] = (d['s'] - d['s'].mean()) / d['s'].std()
d = d.merge(rets, on=['ticker', 'day']) # rets: ticker, day, xret (%)
fit = smf.ols('xret ~ z', data=d).fit(cov_type='cluster',
                                     cov_kws={'groups': pd.factorize(d['day'])[0]})
```

C2 [Proposed]: Critique an AI Answer — Task

- ▣ **Model:** B1, B5 and B7 [Solved]
- ▣ **Conclusion of the AI assistant**
 - ▶ “Accuracy 74.2%, more than twice the 33% of a random guess among three classes: Qwen2.5-7B reads financial sentiment very well.”
 - ▶ “Slope +41.4 bp per standard deviation of the score, $t = +19.07$ (clustered by day): the day’s headlines forecast the stock’s excess return.”
 - ▶ “Qwen2.5 was never trained on these labels or return series, so the 2009–2023 backtest is fully out of sample.”
- ▣ **Question:** the code runs and the numbers are right, yet the answer contains **three** errors typical of sentiment backtests; which?
- ▣ **Tasks**
 - ▶ (a) Find each error: name the line of code or the sentence and explain why it is wrong.
 - ▶ (b) Say how you would detect it without knowing the answer: a class count, a timestamp check, a release date, a memory test.
 - ▶ (c) Write the corrected code or statement, with the numbers it produces.
 - ▶ (d) Log each error in [AI_ERRORS.md](#) with the fields: what the AI answered, what was wrong, how you found out, the correct version.

B2 [Proposed]: Is a 7-Billion-Parameter LLM Better than FinBERT? — Task

- **Model:** B1 [Solved]
- **Question:** does a general open LLM, used zero-shot, beat the fine-tuned FinBERT on headlines neither was trained on?
- **Data**
 - ▶ Twitter Financial News, validation set: $N = 2\,388$ headlines from 2022, each labelled negative, neutral or positive; the human label is treated as the truth
 - ▶ $A = \text{FinBERT}$, $B = \text{Qwen2.5-7B}$ zero-shot with prompt P1; neither used any Twitter label
- **Tasks**
 - (a) Compute each accuracy with a percentile bootstrap 95% CI: resample the headlines 2000 times (seed 0).
 - (b) Compute the difference $B - A$ with a paired bootstrap CI: the same resampled headlines for both classifiers.
 - (c) Count n_{01} , n_{10} ; run the exact McNemar test and $(n_{01} - n_{10})^2 / (n_{01} + n_{10})$ against χ^2_1 (3.84).
- **Report:** the same table, tests and chart as in B1
- **Interpretation:** do the two intervals overlap, does that settle the question, and what would a much larger, slower model still bring?

Submission Checklist and Team Repository

Hand in before the next seminar

- ▣ A1, A3, A7: the derivations and the numbers
- ▣ B1, B5, B7: one table and one figure each, as in the solutions
- ▣ C2: the three errors, how you detected them, the corrected code and your [AI_ERRORS.md](#) entries
- ▣ B2: the table, the tests and the pairs chart
- ▣ One interpretation sentence per exercise

Team repository

- ▣ Choose a text source not covered in class (BVB announcements, Romanian financial news, central-bank statements, crypto posts)
- ▣ Label at least 200 texts with two annotators and report the agreement; evaluate only on a held-out part
- ▣ Log AI use in [AI_USE.md](#) and caught errors in [AI_ERRORS.md](#); extension packs are optional, C1 and C3 are project starts

Extension Packs: What Each One Needs

Pack	Exercises	Prerequisites
1. Measurement error	A2, A6	A1
2. Comparing classifiers	A4, B3 [Solved], B4, B8	A3, B1
3. LLM probabilities	A5 [Solved]	Notation slide, lecture calibration chart
4. Valid inference	B9 [Solved], B10, B11	B5, B7
5. Signals and costs	A8, B6, C1	A7, B5
6. Replication lab	C3	C1, the lecture case study

- Each exercise has its own notebook section; do them in order within a pack
- Useful for the team project: C1 and C3 are possible project starts

A2 [Proposed]: Attenuation for the LM Dictionary — Task

- **Model:** A1 [Solved]
- **Question:** is the weaker same-day LM slope of Lecture 15 (15.5 bp against 39.2 bp for FinBERT) a weaker market reaction, or a worse measurement?
- **Data**
 - ▶ Twitter Financial News, validation set: $N = 2\,388$ headlines from 2022, each labelled negative, neutral or positive; the human label is treated as the truth
 - ▶ Same model and assumption as A1; now $\hat{s} = 1$ when the LM tone of the headline is negative
- **Tasks**
 - (a) Compute π , p , α_0 and α_1 from the LM matrix.
 - (b) Compute λ and $1/\lambda$ with the formula of A1.
 - (c) Compute $\text{Cov}(u, s^*)$: is the LM error classical?
 - (d) Compare λ with FinBERT (A1): which attenuates more, and by what ratio?
- **Report:** the four rates, λ , $1/\lambda$, the ratio

LM confusion matrix, Twitter

True \ predicted	neg	neu	pos	Total
neg	172	166	9	347
neu	247	1 192	127	1 566
pos	45	349	81	475
Total	464	1 707	217	2 388

A6 [Proposed]: Fleiss' κ and One Dawid–Skene Step — Task

- ▣ **Model:** A1 [Solved]
- ▣ **Question:** how much do raters agree beyond chance, and what is the most likely true label of a disputed sentence?
- ▣ **Inputs**
 - ▶ Illustrative votes (made up for the exercise): 6 sentences, $m = 5$ raters each
 - ▶ $P_i = \sum_j n_{ij}(n_{ij} - 1)/[m(m - 1)]$, $\bar{P} = \frac{1}{N} \sum_i P_i$, p_j = share of votes for j ,
 $\bar{P}_e = \sum_j p_j^2$, $\kappa = (\bar{P} - \bar{P}_e)/(1 - \bar{P}_e)$ [Fleiss \(1971\)](#)
- ▣ **Tasks**
 - (a) Compute every P_i and $\bar{P}$.
 - (b) Compute p_j , $\bar{P}_e$ and κ .
 - (c) E-step of [Dawid & Skene \(1979\)](#): each rater gives the true class with probability 0.8 and each other class with 0.1, uniform prior; compute $P(\text{true} = j \mid \text{votes}) \propto \prod_i \pi_{ji}^{n_{ij}}$ for sentences 4 and 5.
 - (d) Compare the posteriors with the vote shares and describe the M-step.
- ▣ **Report:** κ and the two posteriors
- ▣ **Interpretation:** how would you treat the disputed sentences in a test set?

Sentence	neg	neu	pos
1	5	0	0
2	0	5	0
3	0	1	4
4	1	3	1
5	2	3	0
6	0	2	3

A4 [Proposed]: Is "No Difference" a Power Problem? — Task

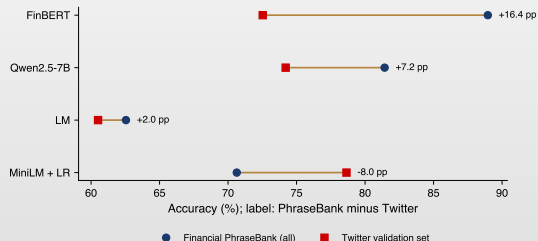
- **Model:** A3 [Solved]
- **Question:** FinBERT and Qwen2.5-7B look tied: is that evidence of equal accuracy, or of a weak test?
- **Inputs**
 - ▶ Twitter validation set, $n = 2\,388$; $A = \text{FinBERT}$, $B = \text{Qwen2.5-7B (prompt P1)}$; $n_{01} = 395$, $n_{10} = 355$
- **Tasks**
 - (a) Compute $\hat{d}$, $\hat{q}$, SE, the 95% CI and z with the formulas of A3.
 - (b) Run the exact McNemar test at 5%.
 - (c) Compute the power at $n = 2\,388$ if the true d is $\hat{d}$, and n_{80} .
 - (d) Use the CI to state which differences the data rule out.
- **Report:** the numbers of (a)–(d)
- **Interpretation:** may you conclude that the two models are equally accurate?

B3 [Solved]: Training-Data Leakage — Task

- ▣ **Question:** how much of FinBERT's accuracy on Financial PhraseBank comes from having been trained on it?
- ▣ **Data**
 - ▶ Financial PhraseBank [Malo et al. \(2014\)](#): 4 846 sentences (2 264 with full agreement); FinBERT was fine-tuned on a split of them (3 101 of 4 845 in [Araci \(2019\)](#))
 - ▶ Twitter validation set, 2 388 headlines unseen by FinBERT; classifiers: FinBERT, Qwen2.5-7B, LM, MiniLM + logistic regression trained on Twitter [Wang et al. \(2020\)](#)
- ▣ **Tasks**
 - (a) Compute accuracy (2 000-draw bootstrap CI) and macro-F1 on PhraseBank with full agreement, all of PhraseBank and Twitter.
 - (b) Compute the drop from all of PhraseBank to Twitter for each classifier.
 - (c) Compute the difference in differences FinBERT minus Qwen2.5-7B with a bootstrap CI (each data set resampled independently).
- ▣ **Report:** a table of accuracies and drops, the difference in differences with its CI
- ▣ **Interpretation:** how do you separate "harder data" from "memory of the training set"?

B3 [Solved]: Solution

Accuracy (%)	PhraseBank, 100% agr.	PhraseBank, all	Twitter	Drop (pp)	Macro-F1, Twitter
FinBERT	97.2	89.0	72.5	16.4	66.8
Qwen2.5-7B	93.6	81.4	74.2	7.2	72.6
LM	65.0	62.5	60.5	2.0	46.2
MiniLM + log. reg.	81.4	70.6	78.6	-8.0	69.9



- (c) $16.4 - 7.2 = 9.2$ pp from unrounded accuracies, bootstrap CI [6.7; 11.9]: excludes 0
- Qwen2.5-7B was not tuned on PhraseBank labels, so its drop measures the harder data; if both would lose the same without overlap (an untested assumption), FinBERT's extra 9.2 points come from overlap
- Interpretation: MiniLM moves the other way because it is trained on Twitter: every model is best at home

B4 [Proposed]: How Many Labels Do You Need? — Task

- ▣ **Model:** B3 [Solved]
- ▣ **Question:** how many labelled headlines does a simple supervised model need to beat FinBERT?
- ▣ **Data**
 - ▶ Twitter Financial News: training set of 9 543 headlines, validation set of 2 388; benchmark FinBERT, 72.5%, no Twitter label used
 - ▶ Logistic regression ($C = 4$) on 384-dimensional MiniLM embeddings; logistic regression ($C = 8$) on TF-IDF features of words and word pairs
- ▣ **Tasks**
 - (a) For $n \in \{100, 200, 400, \dots, 6\,400, 9\,543\}$ draw 10 random training subsets, fit both models and average the validation accuracy.
 - (b) Find the smallest n at which each model beats FinBERT on average.
 - (c) Report the SD of the embedding accuracy over the 10 repetitions at $n = 100$.
- ▣ **Report:** the learning curve and its table
- ▣ **Interpretation:** why do embeddings start higher but grow more slowly?

B8 [Proposed]: Prompt Sensitivity — Task

- **Model:** B1 [Solved]
- **Question:** does the answer of the same LLM change when the same question is worded differently?
- **Data**
 - ▶ Twitter Financial News, validation set: $N = 2\,388$ headlines from 2022, each labelled negative, neutral or positive; the human label is treated as the truth
 - ▶ Qwen2.5-7B and 14B with three prompts: P1 "classify the sentiment . . . for investors", P2 "good, bad or neutral for the stock price", P3 "what is the sentiment of this text"
- **Tasks**
 - (a) Compute accuracy with a 2 000-draw bootstrap CI and the share of "neutral" answers for each prompt and model.
 - (b) Run the exact McNemar test of P1 against P3 for each model.
 - (c) Compute the share of headlines with the same label under all three prompts.
- **Report:** a table and a chart of the three prompts
- **Interpretation:** if you picked the best prompt on this set, what would you be doing?

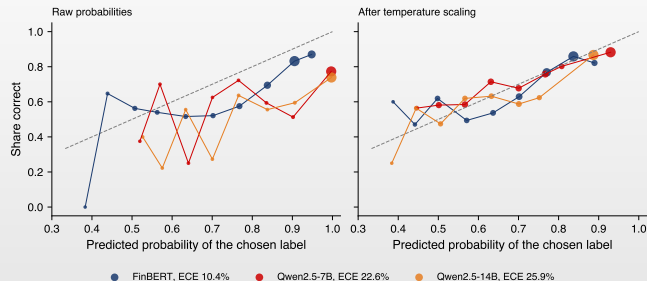
A5 [Solved]: From Label Probabilities to a Score, and the Optimal Temperature — Task

- **Question:** how does the temperature T change an LLM's probabilities, its score and its calibration, and which T fits the data?
- **Inputs**
 - ▶ One Twitter headline labelled **neutral**; Qwen2.5-7B (P1) gives $p_{\text{neg}} = 0.608$, $p_{\text{neu}} = 0.392$, $p_{\text{pos}} = 0.000$ (exactly 1.0×10^{-12})
 - ▶ A random half of the validation set ($n = 1\,194$, seed 15): logits $z_{ik} = \ln p_{ik}$, rescaled $p_{ik}(T) \propto e^{z_{ik}/T}$, loss $\text{NLL}(T) = -\sum_i \ln p_{iy_i}(T)$ Guo et al. (2017)
- **Tasks**
 - (a) Give the label, the net score s and the logit gap $\ln(p_{\text{neu}}/p_{\text{neg}})$ of the headline, then its probabilities and s at $T = 2$.
 - (b) Explain why labels and accuracy do not depend on T , while the Brier score does.
 - (c) Derive $\text{NLL}'(T)$, evaluate the first-order condition at $T = 1$ and solve for T^* (Brent on $[0.5, 50]$).
 - (d) On the other half, compute ECE and Brier before and after scaling.
- **Report:** the label, s , $s(T)$, T^* , ECE and Brier before and after
- **Interpretation:** is the LLM over- or under-confident, and what happens to a rule with a threshold on s ?

A5 [Solved]: Solution (1/2)

- ▣ (a) Largest probability p_{neg} : label negative, wrong; $s = 0.000 - 0.608 = -0.608$; logit gap $\ln(0.392/0.608) = -0.437$
- ▣ At $T = 2$, $p_k(T) \propto \sqrt{p_k}$: $\sqrt{p} = (0.780, 0.626, 1.0 \times 10^{-6})$, sum 1.406, $p(T) = (0.554, 0.446, 0.000)$, $s(T) = -0.554$: same label, score pulled towards 0
- ▣ (b) $z \mapsto z/T$ is increasing for $T > 0$: the order of the logits, hence $\arg \max$ and accuracy, do not change; the probabilities, and with them Brier and log loss, do

A5 [Solved]: Solution (2/2)

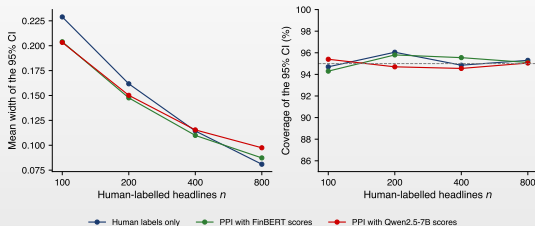


- (c) $\ln p_{iy}(T) = z_{iy}/T - \ln \sum_k e^{z_{ik}/T}$ gives $\text{NLL}'(T) = \frac{1}{T^2} \sum_i (z_{iy_i} - \sum_k p_{ik}(T) z_{ik})$
- At $T = 1$ the average bracket is $-2.47 < 0$, so NLL falls as T grows; the root is $T^* = 8.43$
- (d) ECE falls from 22.6% to 4.3%, Brier from 0.468 to 0.366 (chart: raw and rescaled reliability)
- Interpretation: raw LLM probabilities are heavily overconfident; a threshold on s selects different headlines after rescaling

B9 [Solved]: Prediction-Powered Inference with a Few Human Labels — Task

- **Question:** can a few hundred human labels plus a biased model give a valid and narrower interval for the mean tone of the headlines?
- **Data**
 - ▶ Target θ = mean human net tone $(-1, 0, 1)$ of the 2 388 Twitter validation headlines, which act as the population; f = net score of FinBERT or of Qwen2.5-7B, fixed
 - ▶ Design: n labelled headlines (with Y) and $N - n$ unlabelled ones (only f), drawn independently with replacement
- **Tasks**
 - (a) For one draw with $n = 200$, compute the labels-only estimate $\bar{Y}_n \pm 1.96 s_Y / \sqrt{n}$.
 - (b) Compute PPI [Angelopoulos et al. \(2023\)](#): $\hat{\theta} = \frac{1}{N-n} \sum_{\text{unlab.}} f - \frac{1}{n} \sum_{\text{lab.}} (f - Y)$, $\text{SE} = \sqrt{\hat{V}(f)/(N-n) + \hat{V}(f - Y)/n}$, for both models.
 - (c) For $n = 100, 200, 400, 800$, repeat 2 000 times and record the coverage of θ and the mean CI width.
- **Report:** the one-draw estimates and a coverage and width chart
- **Interpretation:** why is the plain model mean $\frac{1}{N} \sum f$ not an estimate of θ ?

B9 [Solved]: Solution



- True $\theta = 0.054$; plain model means FinBERT -0.014 (wrong sign), Qwen $+0.101$: the bias does not shrink as N grows
- (a) Labels only: $0.010 \pm 1.96 \times 0.042$; (b) PPI-FinBERT: rectifier -0.064 , $\hat{\theta} = 0.054 \pm 1.96 \times 0.036$; PPI-Qwen: rectifier -0.012 , $\hat{\theta} = 0.123 \pm 1.96 \times 0.039$
- (c) $n = 100$: coverage 94.7%, 94.3%, 95.4%, width 0.229, 0.204, 0.203 (labels, PPI-FinBERT, PPI-Qwen); gain for large N : 24% and 27%; at $n = 800$ PPI is wider (0.087 vs 0.081)
- Interpretation: the model mean converges to the wrong number; the labelled sample corrects it, and PPI is narrower only when $\text{Var}(f - Y) < \text{Var}(Y)$ and many texts are unlabelled

B10 [Proposed]: Wild Cluster Bootstrap with 16 Stocks — Task

- **Model:** B5 [Solved]
- **Question:** with only 16 stock clusters, which of the nine slopes of B5 remain significant?
- **Data**
 - ▶ FNSPID panel: 18 282 stock-days with news, 16 US stocks, 27.10.2009–18.12.2023; $z_{i,d}$: daily net score, standardised over the panel; r^e : excess return over SPY (%), from the adjusted daily closes of the course data
 - ▶ The nine regressions of B5 (three scores $\times h = 0, 1, 2$), SE clustered by stock ($G = 16$)
- **Tasks**
 - (a) Compute the p -value of each slope with t_{15} critical values.
 - (b) Run the wild cluster bootstrap with the null imposed [Cameron, Gelbach & Miller \(2008\)](#): $y_i^* = \bar{y} + w_{g(i)}(y_i - \bar{y})$, Rademacher $w_g = \pm 1$, 9 999 draws.
 - (c) Compute t with two-way clustering, $\hat{V} = \hat{V}_{\text{day}} + \hat{V}_{\text{stock}} - \hat{V}_{\text{day} \cap \text{stock}}$.
 - (d) Adjust the nine bootstrap p -values with Holm at 5%.
- **Report:** a table with t , the three p -values and the two-way t
- **Interpretation:** does Qwen at $h = 2$ survive?

B11 [Proposed]: Specification Curve for the Sentiment Strategy — Task

- ▣ **Model:** B7 [Solved] (a randomisation test)
- ▣ **Question:** over all reasonable variants of the strategy, is there more evidence of predictability than chance would give?
- ▣ **Data**
 - ▶ Strategy of Lecture 15: at the close of $d + 1$ take $\text{sign}(s_{i,d})$ in every stock with news on day d , hedged with SPY, closed one day later; four trades on each day with positions; 3 568 trading days
 - ▶ Grid of Lecture 15: three scores $\times$ thresholds $|s| > b$, $b \in \{0, 0.25, 0.5\}$, $\times$ return day $d + 2, \dots, d + 5$: 36 strategies
- ▣ **Tasks**
 - (a) Compute the mean daily return and t_{NW} of each strategy and plot the sorted t -statistics [Simonsohn, Simmons & Nelson \(2020\)](#).
 - (b) Simulate the joint null: flip the signs of the positions in blocks of 10 signal days (dependent wild bootstrap [Shao \(2010\)](#)), the same flips for all 36, 2 000 times.
 - (c) Compute the joint p -values of the number of $t > 1.96$ and of the median t .
- ▣ **Report:** the sorted curve and the two joint p -values
- ▣ **Interpretation:** what would a "best specification" report hide?

A8 [Proposed]: The Qwen Strategy after Costs — Task

- ▣ **Model:** A7 [Solved]
- ▣ **Question:** the Qwen2.5-7B strategy has a statistically positive gross mean: is it tradable?
- ▣ **Inputs**
 - ▶ Strategy of Lecture 15: at the close of $d + 1$ take $\text{sign}(s_{i,d})$ in every stock with news on day d , hedged with SPY, closed one day later; four trades on each day with positions; 3 568 trading days
 - ▶ Qwen2.5-7B signal: $\hat{\mu} = 5.64$ bp, $t_{NW} = 2.26$ ($L = 8$), $a = 0.950$, gross Sharpe ratio 0.63
- ▣ **Tasks**
 - (a) Compute the break-even cost c^* .
 - (b) Compute $\text{SE}(c^*)$ by the delta method and the 95% CI.
 - (c) Recover σ and compute the net Sharpe ratio at 5 bp.
- ▣ **Report:** c^* , SE, CI, net Sharpe ratio
- ▣ **Interpretation:** does a significant gross mean imply a profit after costs?

B6 [Proposed]: Event Study with the Loughran–McDonald Tone — Task

- **Model:** B5 [Solved]
- **Question:** around negative and positive news days, when does the price move, and does the dictionary separate the two groups as well as FinBERT?
- **Data**
 - ▶ FNSPID panel: 18 282 stock-days with news, 16 US stocks, 27.10.2009–18.12.2023; $z_{i,d}$: daily net score, standardised over the panel; r^e : excess return over SPY (%), from the adjusted daily closes of the course data
 - ▶ The daily LM tone is exactly 0 on 48% of stock-days: groups tone < 0 (5 196 stock-days) and tone > 0 (4 288); FinBERT: bottom and top terciles
- **Tasks**
 - (a) For each date, average the excess returns of days $-5, \dots, +5$ over that date's stock-days in each group.
 - (b) Compute the mean over dates of the CAR for $[-5, -1]$, day 0, day 1 and $[+2, +5]$, with t_{NW} .
 - (c) Plot the mean cumulative paths of the four groups.
- **Report:** the table of the four windows and the chart
- **Interpretation:** why is the post-event return positive for both groups [Tetlock \(2007\)](#)?

C1 [Proposed]: Does Headline Sentiment Pay after Costs? — Task

▣ **Model:** A7, B5 [Solved]

▣ **Question:** does a daily strategy on FNSPID headline sentiment earn money after realistic costs?

▣ **Data**

- ▶ Strategy of Lecture 15: at the close of $d + 1$ take $\text{sign}(s_{i,d})$ in every stock with news on day d , hedged with SPY, closed one day later; four trades on each day with positions; 3 568 trading days
- ▶ Positions equally weighted; days without news: return 0; 5 bp per trade; scores FinBERT, Qwen2.5-7B, LM from the saved tables

▣ **Tasks**

- For FinBERT, compute the mean daily gross return, t_{NW} and a moving-block bootstrap 95% CI (blocks of 10 days, 2 000 draws).
- Compute the Sharpe ratio before and after costs, the break-even cost and the Sharpe ratio at 0, 1, 2, 5, 10 bp.
- Repeat for Qwen2.5-7B and LM, and for strong news $|s| > 0.5$ (exploratory); Holm over the three unfiltered strategies, or over all six if you select among them.
- Further step: correct the same-day slope with a random human-labelled sample of stock-days (divide by $\rho(s^*, \hat{s})$, or rectify the OLS moments as in B9).

▣ **Report:** a two-page note with a results table and charts, and a Quantlet folder

▣ **Interpretation:** discuss timing, look-ahead (B7), survivorship of the 16 stocks and crowding [Lopez-Lira & Tang \(2026\)](#)

C3 [Proposed]: Replicating Lopez-Lira and Tang on FNSPID — Task

- ▣ **Model:** C1, B5, B10, B7
- ▣ **Question:** does the published GPT-4 design [Lopez-Lira & Tang \(2026\)](#), run with an open model on the course headlines, find the same reaction and the same drift?
- ▣ **Inputs**
 - ▶ Paper: GPT-4 answers YES, NO or UNKNOWN about each headline; $s = 1, -1, 0$ drives a daily long-short portfolio; overnight drift 0.34% per day, Sharpe ratio 2.97 (Table 1)
 - ▶ Data: FNSPID headlines of the 16 stocks, adjusted daily closes of the course data; first safe return $d + 2$
 - ▶ Model: Qwen2.5-7B-Instruct [Qwen Team \(2024\)](#), temperature 0, the prompt of Section 5 of the paper word for word
- ▣ **Tasks**
 - (a) Score every headline, parse the first line only, map YES/UNKNOWN/NO to 1/0/ - 1; report the shares and invalid answers.
 - (b) Average by stock-day and form the daily portfolio with the conventions of the next slide, earning the returns of day $d + 2$.
 - (c) Report the mean daily return in bp, t_{NW} , the hit rate, the Sharpe ratio, and the shares of days with one leg and with no position.
 - (d) Estimate Eq. (1) at headline level for $h = 0, 2$, with stock and date fixed effects, SE clustered by date and stock and the wild cluster bootstrap p (B10).
 - (e) Recompute the Sharpe ratio net of 5, 10, 20 bp per round trip (Figure 4) and compare with C1 on the same stock-days.
- ▣ **Report:** your numbers next to Tables 1 and 2 of the paper, cumulative returns before and after costs, a Quantlet folder
- ▣ **Interpretation:** which differences from the paper can your data separate, and what does B7 say about the bias?

C3 [Proposed]: Portfolio, Calendar and Cost Conventions

□ Fixed before any run

- ▶ Weights (Table 1): long leg = equal-weighted mean return of the stocks with $s > 0$, short leg = same for $s < 0$, a leg only with at least two stocks; long-short = long minus short (gross exposure 2); if only one leg qualifies, hold it alone (footnote 20)
- ▶ Returns: raw stock returns, as in Table 1; the SPY-hedged version (as in C1) is a course adaptation, reported as a sensitivity: with both legs the hedge cancels, so it matters only on one-leg days
- ▶ Calendar: all trading days, as in C1, with return 0 without a position; mean, t_{NW} and Sharpe ratio over all days; hit rate over days with a position
- ▶ Costs (Figure 4): c bp per round trip on each unit of notional: $2c$ per day with two legs, c with one, plus, in the hedged version, c on the net SPY notional (1 on one-leg days)
- ▶ Regressions (d): raw returns in % per day, as in Table 2; the date fixed effects absorb the market return; the wild cluster bootstrap keeps both fixed effects in the null model

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